

COSMOS PROPRIETARY COMPUTATIONAL ENGINEERING ARCHITECTURE

Architecture pattern, intellectual-property doctrine, RAG knowledge foundation, CAD/topology generation workflow, simulation validation loop, UI/UX strategy, and engineering documentation system

Project: COSMOS - Cryogenic Optimization and Simulation Multiphysics Operating System

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Ownership: Company proprietary and confidential; not a clone of any external proprietary system

Core thesis

COSMOS shall be developed as a proprietary computational engineering compiler for liquid rocket propulsion: requirements become a validated design state, evidence-bound solvers generate engineering parameters, geometry generators compile manufacturable CAD, external simulation tools challenge the result, and a human release gate controls any 3D-printable output.

1. Executive Doctrine

COSMOS is intended to be a desktop-first multiphysics engineering platform for rocket propulsion design, development, verification, and manufacturing preparation. Its purpose is not to imitate a single external tool, but to establish a company-owned computational engineering architecture that uses executable engineering logic, traceable knowledge, deterministic solvers, generated CAD, external simulation evidence, and formal release control.

The software shall work in a similar category to modern computational-engineering systems: the user enters mission and engine requirements, the system converts those requirements into a formal design contract, the engineering kernel calculates a candidate engine, and the geometry kernel produces CAD models that can be exported to specialist analysis and manufacturing systems. COSMOS must remain independent in source code, data schema, solver implementation, user experience, and proprietary design pattern.

- The RAG system is an evidence and traceability layer, not an unchecked design authority.
- The engineering solvers are deterministic, unit-safe, versioned, and independently testable.
- The generated CAD is parametric and evidence-bound before it becomes optimized or printable.
- Topology optimization proposes candidates; clean CAD reconstruction and external analysis decide release readiness.
- Human review and uploaded external results are mandatory before manufacturing release.

1.1 Product Identity

Element	Definition
Product name	COSMOS - Cryogenic Optimization and Simulation Multiphysics Operating System
Primary language	Python for orchestration, scientific computing, data models, RAG, automation, and first-stage CAD generation
Core category	Computational engineering and AI-assisted propulsion design platform
Primary output	Traceable design state, engineering documentation, CAD/CAE packages, analysis comparison records, and manufacturing release packages
Governance stance	Evidence-first, solver-verified, externally reviewed, human released

1.2 Non-Clone Position

COSMOS may operate in the same broad category as other computational engineering systems, but the company shall treat COSMOS as a clean, independent implementation. No proprietary algorithms, source code, data structures, internal user flows, or confidential know-how from external systems shall be copied or reverse engineered. COSMOS shall be designed from first principles, public scientific knowledge, licensed references, company-generated validation data, and original engineering software.

IP caution

This document is an engineering and product-architecture draft, not legal advice. Formal IP assignment, employee/contractor invention agreements, copyright registration strategy, patent review, and trade-secret controls should be reviewed by qualified counsel.

2. COSMOS Proprietary Architecture Pattern

The proposed proprietary pattern is named TRACE-GEN: Traceable Retrieval-Augmented Computational Engineering Generation. TRACE-GEN is the COSMOS operating pattern that links every design output to a controlled requirement, verified equation, unit model, source reference, solver version, CAD feature, analysis result, and release decision.

TRACE-GEN Stage	Purpose	Primary Artifacts
T - Technical Design Contract	Convert user inputs into formal requirements, constraints, assumptions, and acceptance criteria.	Design contract, requirement IDs, operating envelope
R - Retrieval and Reference Binding	Retrieve approved equations, material data, propellant data, correlations, and manufacturing rules.	Evidence packets, citations, validity ranges
A - Analytical Synthesis	Run deterministic engineering solvers and derive a coherent preliminary engine state.	Sizing results, margins, warnings
C - CAD and Geometry Compilation	Generate parametric and implicit geometry from engineering state.	Feature graph, STEP/Parasolid, mesh seeds

TRACE-GEN Stage	Purpose	Primary Artifacts
E - External Evaluation	Export to independent CAE tools and import solver results for comparison.	CFD/FEA/FEM/dynamics cases, result maps
G - Governance Review	Compare evidence and obtain human review before release.	Review board, signoff record, issue log
E - Engineering Documentation	Generate traceable reports and design basis documents.	Calculation package, design report, release note
N - Next-Design Learning	Use approved result deltas and test data to refine models without corrupting source evidence.	Model calibration records, validation datasets

2.1 Unique COSMOS Differentiators

- Evidence-bound geometry: every CAD feature links to a requirement, solver output, equation, material rule, or manufacturing constraint.
- Design decision ledger: all automated and human decisions are recorded with timestamp, model version, reviewer, source evidence, and acceptance rationale.
- Solver disagreement cockpit: COSMOS predictions, Fluent/OpenFOAM/Nastran/Ansys/NX results, and manual reviewer comments are shown side by side.
- Manufacturing release passport: a design cannot export a print-release package without proof of geometry, mesh, thermal, structural, CFD, and human review gates.
- Equation validity guardian: equations retrieved by RAG are checked for units, dimensions, operating range, status, and source authority before solver use.
- Topology reconstruction discipline: optimized forms are converted back into parametric or implicit engineering geometry before final validation.
- Design lineage replay: a released engine can be regenerated from requirements, data versions, solver versions, and CAD generator versions.
- AI role separation: AI assists search, explanation, planning, comparison, and documentation; deterministic software owns calculations and geometry generation.

2.2 Architectural Non-Negotiables

- No undocumented equation shall enter production solvers.
- No RAG answer shall be used as a direct numerical authority without validation and source binding.
- No STL/3MF export shall be treated as release-ready unless the design passport is complete.
- No topology optimization output shall bypass clean geometry reconstruction.
- No external simulation result shall be accepted without solver version, mesh metadata, boundary conditions, convergence data, and load-case identity.
- No manufacturing package shall be released without human signoff.

3. Requirements-to-Release Workflow

```
User Requirements
-> Design Contract
  -> Evidence Retrieval and Source Binding
    -> Deterministic Sizing and Multiphysics Synthesis
      -> Parametric CAD Generation
        -> Topology / Shape Optimization Candidates
          -> Clean CAD Reconstruction
            -> External CFD / FEA / FEM / Dynamics Validation
              -> COSMOS Comparison Cockpit
                -> Human Review Gate
                  -> Manufacturing Release Passport
                    -> STEP / CAE / 3MF-STL / Engineering Report Package
```

The initial user inputs may include propellant pair, target thrust, chamber pressure, burn time, feed architecture, cooling strategy, manufacturing method, material preferences, altitude condition, safety factors, and packaging limits. COSMOS shall immediately convert these into a design contract rather than passing raw values directly into solvers.

Input Class	Examples	Validation Requirement
Mission and operating point	Thrust, burn time, altitude, duty cycle, restart count	Finite values, units, completeness, intended regime
Propulsion configuration	Pressure-fed, pump-fed, gas-generator, expander, staged-combustion future options	Supported cycle and maturity status
Propellants	LOX/LCH4, LOX/LH2, LOX/RP-1, future monopropellant or storable options	Traceable property database and CEA species mapping
Materials and manufacturing	Inconel, copper alloys, GRCo family, additive process, machining limits	Material source, thermal/structural limits, process capability
Review strategy	Internal validation, external Fluent/OpenFOAM/Nastran/Ansys review, test correlation	Reviewer identity and required evidence package

4. Layered Software Architecture

The existing COSMOS layered architecture remains correct and should be extended with knowledge, CAD generation, optimization, simulation adapters, documentation, and release-governance layers. Dependencies should still point downward: UI calls backend APIs; backend orchestrates systems; systems combine physics models; physics depends only on core utilities and approved scientific libraries.

Layer	Responsibilities	Forbidden Responsibilities
Core	Units, constants, validation, logging, configuration, exceptions, numerical safety primitives	Engineering equations, GUI, workflow policy
Knowledge Foundation	References, documents, equations, variables, units, constants, materials, propellants, assumptions, validity ranges	Unverified AI-generated equations, direct solver execution

Layer	Responsibilities	Forbidden Responsibilities
Physics	Thermochemistry, gas dynamics, heat transfer, fluids, materials physics, structural formulas	Application workflows, CAD decisions, UI
Systems	Injector, cooling, chamber/nozzle, igniter, feed-system, structure, reliability assemblies	GUI and persistence internals
Backend	Solver orchestration, optimization runs, dependency scheduling, project state, API boundaries	Raw equation definitions without source traceability
CAD/Geometry	Parametric models, implicit geometry, feature graph, topology reconstruction, export packages	Unsupported manufacturing release decisions
Simulation Adapters	Ansys, Fluent, OpenFOAM, Nastran, NX, Abaqus, Simcenter, Adams, Simulink, Dymola integration	Silently modifying design intent
GUI/UX	Input capture, workflow navigation, results display, comparison cockpit, review boards	Performing calculations directly
Documentation	Design basis, calculation packages, analysis reports, release passports, audit logs	Untraceable or manually detached reports

4.1 Primary Python Technology Stack

Domain	Recommended Primary Tools	Purpose
Core language	Python 3.11+ initially; C++/Rust optional later	Fast iteration, scientific ecosystem, automation, integration
Data models	Pydantic or dataclasses with strict validators	Schema validation, serialization, traceability
Numerics	NumPy, SciPy, SymPy, Pint	Solvers, optimization, symbolic checks, unit safety
MDO	OpenMDAO; Dakota/optiSLang adapters	Multidisciplinary design optimization and uncertainty
CAD	CadQuery/OCP/OpenCASCADE; optional FreeCAD bridge; later C++ geometry kernels if required	STEP/Parasolid-style parametric generation and solid modeling
Mesh/visualization	Gmsh, meshio, VTK/PyVista	Mesh generation, conversion, result visualization
RAG	Vector database plus keyword index plus knowledge graph	Evidence retrieval with citations and relationships
Storage	SQLite/DuckDB for local, PostgreSQL for team/enterprise, HDF5/Zarr for large results	Project data, result arrays, audit trails
Desktop UI	PySide6/Qt with VTK/Qt 3D views; optional local web UI for advanced dashboards	Professional engineering workbench
Internal API	FastAPI or typed service layer	Decoupled UI/backend and future cloud/automation support

5. Knowledge Foundation and RAG Architecture

The knowledge folder is the source of the AI/RAG system. Its mission is to extract, normalize, govern, retrieve, and explain the mathematics, physics, chemistry, materials science, manufacturing knowledge, and engineering methods required to design and validate rocket engines. It must behave as a controlled technical-data system, not as a free-form chatbot memory.

Knowledge Entity	Required Fields	Engineering Role
Reference	Source ID, authors, title, publisher, year, DOI/URL/ISBN, status, license, confidence	Establishes authority and permitted use
Document	Document ID, source reference, content hash, version, section map, security level, import pipeline	Machine-readable source text
Equation	Expression, variables, units, dimensions, assumptions, validity range, source page/section, verification tests	Reusable solver knowledge
Variable	Symbol, name, unit, dimension, bounds, role, finite/zero/negative policy, source	Typed engineering quantity
Unit	Unit ID, symbol, dimension vector, conversion, offset, source, status	Prevents unit drift
Constant	Value, uncertainty, CODATA/source version, exactness, units, dimensions	Validated numerical constants
Material	Thermal, mechanical, fatigue, compatibility, AM parameters, source, allowables	Design and manufacturing constraints
Propellant	Species mapping, composition, density, thermophysical properties, storage conditions, source	Thermochemistry and feed-system inputs
Subsystem	Functional role, interfaces, constraints, related equations, test data	Links knowledge to engine architecture
Validation Dataset	Experiment/test/simulation source, conditions, measured outputs, uncertainty	Model verification and calibration

5.1 RAG Retrieval Rules

- RAG responses must cite approved references and document sections.
- Equation retrieval must include validity range, variables, units, dimensional signature, assumptions, and verification status.
- Conflicting sources must be surfaced as conflicts, not averaged silently.
- The AI may propose candidate design logic, but production solvers must use approved equation objects and test coverage.
- Every retrieved artifact must be access-controlled according to its license and company security level.
- Generated summaries must never overwrite source truth; they are derived artifacts linked to source hashes.

5.2 Ingestion and Approval Pipeline

1. Acquire source: textbook, NASA report, NIST data, standard, internal test report, supplier data, or validated simulation package.
2. Capture license and access rights before ingestion.
3. Convert to machine-readable text with page, section, table, and figure preservation where possible.
4. Extract candidate equations, variables, material data, propellant data, and assumptions.
5. Run unit and dimensional checks on every candidate equation.
6. Assign confidence, source status, and domain-owner review state.
7. Store immutable source document, derived structured objects, embeddings, keyword indexes, and graph relationships.
8. Promote only reviewed objects into solver-eligible status.

5.3 Knowledge Graph Relationships

```
Reference -> Document -> Section -> Equation -> Variables -> Units -> Dimensions
Equation -> Assumptions -> Validity Range -> Verification Test -> Solver Function
Material -> Manufacturing Process -> Allowables -> Structural Analysis Case
Propellant -> Thermochemistry Model -> Combustion Solver -> Cooling Solver
CAD Feature -> Solver Output -> Requirement -> Evidence Packet -> Human Review
```

6. Engineering Computation Kernel

The computation kernel is the deterministic heart of COSMOS. It shall transform a design contract into a coherent design state by using versioned, tested, traceable physics and systems solvers. The AI/RAG layer can explain or retrieve; it cannot replace solver validation.

Solver Family	Responsibilities	Early Maturity Target
Thermochemistry	CEA/RocketCEA/Cantera integration, propellant properties, mixture ratio sweeps, combustion products	Validated external-engine adapter
Performance	Thrust, Isp, c-star, expansion ratio, mass flow, pressure ratios	Preliminary design capability
Chamber and nozzle	Chamber volume, contraction ratio, throat, bell contour, expansion, contour families	Parametric geometry-ready outputs
Injector	Injector type selection, orifice sizing, pressure drop, spray assumptions, face layout	Initial liquid/liquid injector generator
Cooling	Regenerative channel sizing, heat flux correlation, pressure drop, wall temperature estimates	Conservative steady-state model
Structure	Hoop stress, thermal stress, buckling, margins, fatigue pre-checks	Pre-FEA screening
Ignition	Igniter sizing basis, energy requirement, interface geometry	Conceptual generator with review gate
Feed dynamics	Tank pressure, line losses, valve/transient model hooks	Simscape/Modelica adapter initially
Reliability	FMEA, hazards, margin tracking, failure-mode catalog	Design review support
Optimization	Parametric MDO, topology candidate loops, uncertainty sweeps	OpenMDAO-first orchestration

6.1 Design State Object

A central EngineDesign object shall be the source of truth for a generated engine. It should be serializable, versioned, hashable, and replayable. It must include requirements, assumptions, solver inputs, solver outputs, CAD parameters, analysis packages, review status, and manufacturing-release evidence.

- Requirements: thrust, burn time, chamber pressure, propellant pair, mission point, safety factors.
- Design choices: cycle, mixture ratio, cooling method, injector family, material system, manufacturing process.
- Solver outputs: mass flow, throat area, expansion ratio, wall heat flux, channel dimensions, pressure drops, stresses.
- Geometry parameters: chamber length, radii, throat contour, nozzle control points, channel path, injector layout.
- Analysis records: mesh metadata, load cases, boundary conditions, convergence, result fields, comparison deltas.
- Release records: reviewer identity, open issues, waivers, signoff status, exported files, checksums.

7. CAD and Geometry Generation

COSMOS should generate parametric CAD first. Direct mesh generation is not enough for engineering design control. A parametric model preserves design intent, lets users export to Siemens NX, Ansys, Nastran, Fluent, OpenFOAM, and manufacturing systems, and enables future regeneration when requirements or solver models change.

Geometry Object	Generation Strategy	Primary Export
Thrust chamber	Parametric chamber barrel, contraction, throat, nozzle contour, wall thickness rules	STEP, Parasolid if licensed, neutral mesh
Nozzle	Bell, conical, contour families; shape optimization around control points	STEP, IGES, CFD surface mesh
Regen channels	Parametric helical/axial channels, manifolds, rib thickness, pressure-drop constraints	STEP, analysis mesh, manufacturing report
Injector head	Pattern generator for element families, manifolds, faceplate, pressure-drop metadata	STEP, hole table, manufacturing drawing package
Igniter	Parameterized igniter chamber, ports, interfaces, feed lines, mounting	STEP, analysis-ready solid
Manifolds	Implicit/parametric flow distribution geometry, topology-assisted reinforcement	STEP, CFD/FEA cases
Mounts and interfaces	Envelope-aware brackets, flanges, seals, fasteners, instrument ports	STEP, Nastran/Abaqus structural model

7.1 Evidence-Bound CAD Feature Graph

Every CAD feature should carry metadata. For example, a throat radius feature should know the requirement it satisfies, the solver equation that produced it, the material and wall-temperature limit it must respect, the analysis cases that evaluated it, and the reviewer who accepted or rejected it. This feature-level traceability is one of the strongest unique COSMOS differentiators.

```
CADFeature(  
  feature_id='nozzle.throat.blend.radius',  
  generated_by='geometry.nozzle.v0_2',  
  source_requirements=['REQ-THRUST-001', 'REQ-PC-001'],  
  solver_outputs=['SOL-GASDYN-THROAT-AREA-004'],  
  evidence=['EQ-CHOKED-FLOW-001', 'MAT-INCONEL718-ALLOWABLE-003'],  
  analysis_cases=['FEA-THERMAL-021', 'CFD-NOZZLE-009'],  
  review_status='human_review_required'  
)
```

8. Topology and Shape Optimization

Topology optimization shall be implemented as a candidate-generation and design-improvement layer, not as a direct manufacturing authority. It is most valuable for support structures, jackets, manifolds, mounts, brackets, injector structural supports, and additive-manufacturing lightweighting. Nozzle contours, cooling-channel paths, throat blends, and wall thickness should primarily use parametric or shape optimization because manufacturability and fluid/thermal constraints are tightly coupled.

Optimization Type	Best Use	Release Rule
Topology optimization	Mass reduction, stiffness, thermal paths, structural supports, manifolds, AM lattices	Must be reconstructed into clean CAD and reanalyzed
Shape optimization	Nozzle contour, throat blend, injector passages, channel dimensions	Must preserve constraints and regenerate parametric geometry
Parametric optimization	Mixture ratio, expansion ratio, chamber length, channel count, wall thickness	Runs inside design-state/MDO workflow
Lattice/implicit optimization	AM thermal/structural tailoring, jacket reinforcement, support skins	Requires process-specific manufacturability checks
Robust optimization	Uncertainty margins for material, thermal, pressure, manufacturing tolerance	Must report sensitivity and confidence intervals

8.1 Topology Loop

9. Generate conservative parametric baseline geometry.
10. Define preserve regions, loads, boundary conditions, manufacturing constraints, thermal paths, and no-go volumes.
11. Run topology or shape optimization with explicit objective and constraints.
12. Convert raw output into clean parametric or implicit CAD.
13. Run geometry checks, wall-thickness checks, AM overhang/support checks, and mesh-quality checks.
14. Re-run external CFD/FEA/thermal analysis.
15. Compare optimized result against baseline in COSMOS.
16. Require human review before release.

9. External Simulation and Dynamic Validation

COSMOS should be solver-neutral and export complete analysis packages. External software results should be uploaded back into COSMOS for side-by-side comparison against internal predictions. This secondary review loop is essential for credibility before any 3D-printing release.

Validation Domain	Recommended Tools	COSMOS Integration
CAD and manufacturing review	Siemens NX, FreeCAD, nTop, CAM systems	STEP/Parasolid import/export, feature metadata, manufacturing notes
CFD and combustion flow	Ansys Fluent, OpenFOAM, STAR-CCM+, SU2	Mesh, boundary conditions, residuals, fields, heat flux, pressure drop
Structural FEA/FEM	Ansys Mechanical/Workbench, MSC Nastran, Abaqus	Loads, constraints, thermal maps, stress, strain, safety factor, modal data
Explicit/transient dynamics	LS-DYNA, Abaqus Explicit	Shock, burst, impact, severe transient load cases
Multibody dynamics	MSC Adams, Simcenter 3D Motion	Gimbal, actuator, valve, mechanism, mounting loads
System dynamics	MATLAB Simulink/Simscape, Dymola/Modelica	Feed-system transients, pressurization, valve dynamics, control loops
Optimization/UQ	OpenMDAO, Dakota, optiSLang, modeFRONTIER	MDO, sensitivity, uncertainty, design-space exploration

9.1 Imported Result Package Requirements

- External solver name, version, license environment, and date.
- Geometry file checksum and design-state version.
- Mesh count, mesh-quality metrics, units, coordinate system, and named selections.
- Boundary conditions, material model, turbulence/combustion/thermal model, load case, and solver settings.
- Residuals, convergence criteria, failed warnings, and reviewer notes.
- Primary outputs: temperatures, pressures, stresses, deformation, modal frequencies, heat flux, pressure drop, margin.
- Difference table comparing external results with COSMOS predictions.

9.2 Dynamic Simulation Recommendation

For dynamic simulation, COSMOS should start with system-level transients and mechanism dynamics before attempting high-fidelity transient combustion. Simulink/Simscape and Modelica/Dymola are strong for feed-system dynamics, pressurization, and controls. Adams or Simcenter Motion are strong for gimbals, valves, actuators, and moving assemblies. LS-DYNA or Abaqus Explicit should be reserved for severe transient structural events and nonlinear dynamics.

10. UI/UX Architecture

The UI should feel like a professional engineering workbench rather than a chatbot or calculator. The user should see requirements, evidence, geometry, solver status, external validation, and release readiness as one coherent digital thread.

Workspace	Purpose	Key UI Elements
Dashboard	Project overview and release readiness	Design status, open risks, latest run, review passport
Design Contract	Requirements and constraints	Validated forms, unit-aware inputs, requirement IDs
Evidence Browser	RAG and knowledge foundation	Citations, equations, assumptions, validity ranges
Sizing and Solvers	Engineering computation	Run graph, solver logs, convergence, warnings, margins
Geometry Studio	CAD generation and topology candidates	3D viewer, feature tree, parametric controls, export panel
Simulation Hub	External case management	Tool adapters, case packages, upload results, mesh metadata
Comparison Cockpit	Side-by-side validation	COSMOS vs external result tables, plots, deltas, pass/fail
Documentation Center	Engineering reports and design basis	Templates, generated docs, versioned outputs
Release Gate	Human review and manufacturing release	Checklist, issue log, signoff, export controls

10.1 UI Principles

- Desktop-first, local-first, professional engineering layout.
- Dense but readable workspaces; avoid marketing-style screens in operational views.
- All numerical inputs are unit-aware and validation-aware.
- Every output has an evidence or computation trail accessible from the UI.
- The AI assistant appears as an engineering copilot and review explainer, not as the hidden authority.
- 3D geometry, plots, tables, solver logs, and citations should be visible together when relevant.
- Release readiness must be visible as a formal state, not implied by successful geometry generation.

11. Engineering Documentation System

COSMOS shall generate engineering documentation automatically and store it as traceable design evidence. Documentation is not a side output; it is part of the design authority. Every report should be generated from the design state, source evidence, solver outputs, CAD metadata, external analysis results, and review records.

Document Type	Contents	Release Role
Design Basis Document	Requirements, assumptions, selected architecture, propellants, materials, validity ranges	Defines why the engine exists and what it must satisfy
Calculation Package	Equations, variables, constants, solver versions, numerical outputs, margins	Auditable engineering computation record

Document Type	Contents	Release Role
CAD Generation Report	Feature tree, parametric values, generator version, geometry checks, export files	Shows how geometry was produced
Topology Optimization Report	Objectives, constraints, preserve regions, candidate comparison, reconstruction notes	Prevents raw topology output from becoming unreviewed design truth
CFD Report	Mesh, BCs, solver model, residuals, fields, comparison deltas	Fluid/thermal validation evidence
FEA/FEM Report	Loads, constraints, material model, stress/strain/thermal results, margins	Structural validation evidence
Manufacturing Release Passport	All required gates, signoffs, file checksums, open waivers, print/process notes	Final release-control artifact
Change Impact Report	Before/after requirements, geometry, solver outputs, analysis deltas	Controls revisions and regression risk

11.1 Documentation Storage and Traceability

- Every generated document receives a document ID, version, content hash, source design-state hash, and authoring pipeline version.
- Documents remain immutable after release; revisions create new versions.
- Reports shall be exportable as DOCX, PDF, Markdown, JSON evidence bundles, and project archive records.
- Every report section should be traceable to source data or generated computation.
- External analysis reports uploaded by reviewers should be stored beside generated COSMOS reports and compared in the UI.

12. Data, Persistence, and Digital Thread

COSMOS needs a digital-thread storage model that can operate locally but later scale to team and enterprise use. The first implementation can use SQLite/DuckDB plus file-based artifact storage; the architecture should leave a path to PostgreSQL, object storage, and vector/graph services.

Data Store	Recommended Initial Choice	Future Enterprise Option
Project metadata	SQLite	PostgreSQL
Analytical tabular results	DuckDB/SQLite	PostgreSQL/DuckDB server
Large arrays and fields	HDF5/Zarr	Object storage plus metadata DB
CAD artifacts	Versioned files with checksums	PDM/PLM integration
RAG vectors	FAISS/Qdrant local	Qdrant/pgvector managed service
Knowledge graph	NetworkX/SQLite initially	Neo4j/PostgreSQL graph extensions
Generated documents	File store plus document table	Document management integration
Audit logs	Append-only JSONL/SQLite	Immutable audit service

13. Security, IP, and Compliance Controls

COSMOS should be treated as company intellectual property from the beginning. Source code, schemas, solver logic, CAD generators, topology workflows, curated datasets, generated reports, UI workflows, and validation pipelines are proprietary company assets unless explicitly licensed otherwise.

Control Area	Requirement
Ownership	All contributors must have clear invention assignment, contribution, and confidentiality terms.
Third-party code	Track licenses and never mix incompatible code into proprietary modules.
Reference data	Store license, source, access rights, and allowed-use status for every ingested source.
Trade secrets	Protect solver workflows, architecture pattern, model calibration data, and CAD generator logic.
Access control	Role-based access for source data, projects, release packages, and export functions.
Auditability	Record who generated, changed, reviewed, exported, or released every design artifact.
Export control	Flag propulsion design data for jurisdiction-specific legal review before external sharing.
Clean-room posture	Do not reverse-engineer external proprietary computational-engineering products.

14. API and Interoperability Strategy

COSMOS shall interoperate with major engineering tools through neutral formats and adapter packages. The goal is not to replace every high-fidelity solver immediately, but to generate complete, traceable case packages and to ingest their results.

Artifact	Preferred Formats
CAD solids	STEP AP242, STEP AP214, IGES, Parasolid where licensed
Mesh	CGNS, Gmsh MSH, SU2, VTK/VTU, Nastran BDF, Abaqus INP, Ansys CDB
OpenFOAM	OpenFOAM case folders with boundary, system, constant directories
CFD/FEA results	VTK/VTU, CGNS, CSV summaries, HDF5/Zarr field data, JSON metadata
Print preparation	3MF preferred, STL only as final mesh export after release gate
Documentation	DOCX, PDF, Markdown, JSON evidence bundle
Project archive	COSMOS project package with manifest, checksums, versions, and dependency lock

15. Quality, Verification, and Scientific Credibility

The software cannot become world-class unless its validation is treated as a first-class subsystem. COSMOS must distinguish between conceptual design, preliminary design, analysis-ready geometry, externally

validated design, and manufacturing release. Passing a calculation is not the same as being safe to manufacture.

Quality Gate	Required Evidence
Code quality	Compile, lint, type-check, unit tests, integration tests, coverage trend
Numerical safety	Finite-number checks, units, dimensions, bounds, operating ranges
Equation verification	Known examples, source reproduction, independent derivation checks
Solver validation	Benchmark data, external solver comparison, uncertainty ranges
CAD validation	Watertight solids, feature integrity, wall thickness, interference, manufacturability
Mesh validation	Mesh quality, named boundaries, unit consistency, reproducible case generation
External review	Imported results, comparison deltas, reviewer comments, accepted waivers
Release control	Open issue closure, human signoff, artifact checksums, final export record

15.1 Release State Model

State	Meaning
DRAFT	Requirements or knowledge are incomplete.
CONCEPTUAL	Preliminary sizing exists, but geometry or analysis is incomplete.
ANALYSIS_READY	Geometry and case packages are ready for external analysis.
ANALYSIS_RUNNING	External or internal simulations are active.
ANALYSIS_FAILED	One or more validation cases failed or did not converge.
REVIEW_REQUIRED	Automated checks passed enough for human review.
RELEASE_BLOCKED	Open risks, missing evidence, or failed checks prevent manufacturing export.
MANUFACTURING_RELEASED	Human signoff completed and release passport generated.

16. Development Roadmap

The roadmap should avoid trying to build the full vision at once. A credible path starts with one narrow engine family, one propellant pair, one parametric geometry generator, one external simulation round trip, and one documentation package.

Phase	Scope	Exit Criteria
0 - Foundation hardening	Fix package imports, compile errors, validation gaps, project tooling, canonical data	Full test suite, compileall, ruff, mypy baseline pass

Phase	Scope	Exit Criteria
1 - Knowledge/RAG MVP	Reference/document/equation ingestion, vector+keyword search, citation answers	Approved equations can be retrieved with source and validity data
2 - Preliminary engine sizing	Propellant pair, thrust, Pc, burn time, chamber/nozzle sizing	Deterministic design state and calculation report
3 - Parametric CAD MVP	Chamber, throat, nozzle, simple injector placeholder, STEP export	Regeneratable CAD with feature metadata
4 - External analysis round trip	OpenFOAM/Fluent and Nastran/Ansys package export/import	Comparison cockpit with uploaded results
5 - Regen cooling and injector depth	Channel geometry, pressure drop, heat flux, injector face layout	Analysis-ready chamber/injector assemblies
6 - Optimization layer	OpenMDAO parametric optimization and topology candidate workflow	Baseline vs optimized comparison and reconstruction report
7 - Release passport	Human review workflow, documentation center, manufacturing export controls	No print export without completed release gate
8 - Enterprise maturity	Team database, access control, plugin SDK, validation dataset management	Multi-user digital thread and audit capability

17. Immediate Repo-Level Actions

- Fix knowledge package path mismatch between repository tests and implementation.
- Repair or quarantine the non-compiling Quantity model before it enters the active RAG schema.
- Promote one propellant database as canonical and ensure the default loader points to it.
- Add pyproject.toml, dependency lock, formatter/linter config, and CI commands.
- Implement shared finite numeric validation and apply it across propellants, variables, units, constants, and materials.
- Remove debug print statements from models and prevent import-time side effects in configuration/settings.
- Create first design-state schema before writing more CAD or solver features.

18. Appendix A - Initial COSMOS Module Map

cosmos/	
core/	units, constants, validation, config, logging
knowledge/	RAG schemas, ingestion, graph, retrieval, approval
physics/	equations and scientific models only
systems/	chamber, nozzle, injector, cooling, igniter, feed structure and reliability assemblies
backend/	orchestration, workflows, APIs, project state
geometry/	CAD generators, feature graph, export adapters
optimization/	OpenMDAO, topology/shape loops, UQ, sensitivity
simulation_adapters/	Fluent, OpenFOAM, Nastran, Ansys, Abaqus, NX, Simulink
validation/	benchmarks, external comparison, release gates
documentation/	report templates, design basis, release passport
gui/	Qt workbench, 3D viewer, comparison cockpit
databases/	project, knowledge, material, propellant, result stores
tests/	unit, integration, regression, validation datasets

19. Appendix B - Human Review Checklist

- Design contract complete and approved.
- All equations and material/property data are source-bound and in approved status.
- All solver inputs are finite, unit-valid, dimension-valid, and within validity range.
- CAD solids pass geometry integrity and manufacturability checks.
- External CFD result package uploaded and compared.
- External FEA/FEM result package uploaded and compared.
- Thermal and cooling margin reviewed.
- Structural margin, modal/vibration risk, and fatigue assumptions reviewed.
- Topology reconstruction, if used, was reanalyzed after CAD cleanup.
- Open risks, assumptions, and waivers are explicitly accepted or closed.
- Manufacturing process, material, inspection, and test-plan notes are attached.
- Human reviewer signs release passport before 3MF/STL manufacturing export.

20. Appendix C - Design Input/Output Contract

Category	Minimum Fields
User inputs	Propellant pair, thrust, chamber pressure, burn time, environment, feed architecture, cooling method, material/process, safety factors
Computed outputs	Mass flow, mixture ratio, throat area, chamber dimensions, expansion ratio, nozzle contour, cooling channels, injector layout, wall thickness
CAD outputs	Chamber/nozzle assembly, regen jacket/channels, injector head, igniter, manifolds, mounts, interfaces
CAE outputs	CFD case, FEA/FEM case, thermal case, dynamics/system model package, mesh files, result metadata
Documentation outputs	Design basis, calculation package, CAD report, analysis comparison, review passport, change report

Category	Minimum Fields
Manufacturing outputs	STEP/Parasolid, drawings/tables, 3MF/STL after release, inspection notes, process assumptions

21. Closing Direction

COSMOS should be built as a proprietary evidence-driven computational engineering system. Its unique value should come from the integration of traceable RAG knowledge, deterministic engineering solvers, feature-level CAD provenance, topology-aware design improvement, external solver comparison, generated engineering documentation, and controlled human release. That combination is the company-owned pattern. The immediate task is to harden the foundation, then build one narrow end-to-end engine workflow that proves the TRACE-GEN loop from requirements to reviewed CAD and documentation.